



PROJECT NAME: POCKET SMART AI

TEAM ID:

A PROJECT REPORT

Submitted by

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POCKET SMART AI

FINAL PROJECT DOCUMENTATION

1. ABSTRACT

Pocket Smart AI is a smart budget and recommendation assistant designed to help users plan home interiors, parties and jewelry requirements according to their preferences and budget. The application combines a web interface, FastAPI backend, database support and AI-assisted recommendation functionality.

2. INTRODUCTION

Planning a purchase, event or personal requirement within a fixed budget can be difficult because users need to consider several requirements at the same time. Pocket Smart AI provides a centralized web application where users can enter their requirements and budget and receive recommendation results through a simple interface.

3. PROBLEM STATEMENT

Users may spend considerable time deciding suitable options for home planning, party planning and jewelry requirements. The proposed system organizes user inputs and budget information and provides a structured recommendation workflow.

4. OBJECTIVES

- *Provide a simple web interface for budget-based planning.*
- *Support home, party and jewelry planning.*
- *Provide user registration and login.*
- *Accept user requirements, preferences and budget information.*
- *Generate recommendation results.*
- *Store and display recommendation history.*
- *Combine application services with AI-assisted recommendation processing.*

5. PROPOSED SYSTEM

Pocket Smart AI provides a web-based planning assistant. After authentication, a user selects a planning category, enters requirements and budget information, and generates a recommendation. Previous recommendations can be accessed through the history section.

6. FUNCTIONAL REQUIREMENTS

- *User registration and login*
- *Dashboard navigation*
- *Home Planner*
- *Party Planner*
- *Jewelry Planner*

- *Recommendation generation*
- *Recommendation history*
- *AI-assisted recommendation processing*
- *Database storage*

7. NON-FUNCTIONAL REQUIREMENTS

- *Simple and responsive user interface*
- *Secure handling of application configuration*
- *Reliable backend API*
- *Maintainable modular project structure*
- *Error and fallback handling*
- *Easy local installation and execution*

8. TECHNOLOGY STACK

- *Frontend: HTML, CSS, JavaScript, Jinja2 templates*
- *Backend: Python, FastAPI, Uvicorn*
- *Database: SQLite, SQLAlchemy*
- *AI: Google Gemini API, google-genai SDK*
- *Development: Visual Studio Code, Git, GitHub*

9. SYSTEM MODULES

Authentication Module - registration, login and session-based access.

Dashboard Module - navigation to the planning modules and recent activity.

Home Planner - home-related requirements and budget information.

Party Planner - event-related requirements, budget and preferences.

Jewelry Planner - jewelry-related requirements, occasion and budget.

Recommendation Module - processes planner information and displays recommendation results.

History Module - displays previously generated recommendations.

10. SYSTEM ARCHITECTURE

User → Web Browser / Frontend → FastAPI Application → Routers and Services → Database + AI Recommendation Service → Recommendation Result → User Interface.

The application follows a modular structure separating authentication, page routing, planner APIs, catalog services, recommendation services and AI integration.

11. DATABASE

The application uses SQLite for local persistence and SQLAlchemy for database interaction. User and recommendation-

related information can be maintained through the application database.

12. AI INTEGRATION

The application uses the Google Gemini API through the google-genai SDK for AI-assisted recommendation generation. Planner information is prepared as input for the recommendation process and the resulting structured information is displayed through the application.

The application also includes fallback handling when the external AI service is temporarily unavailable.

13. USER WORKFLOW

- 1. Open Pocket Smart AI.*
- 2. Register or log in.*
- 3. Open the dashboard.*
- 4. Select Home, Party or Jewelry Planner.*
- 5. Enter requirements and budget.*
- 6. Generate a recommendation.*
- 7. View the recommendation result.*
- 8. Access previous results through History.*

14. TESTING

- Registration and login*
- Dashboard navigation*
- Home Planner*
- Party Planner*
- Jewelry Planner*
- Recommendation API*
- Database operations*
- History*
- Frontend pages*
- Error and fallback handling*
- The application was run locally using FastAPI/Uvicorn and the major user flows were checked through the browser.*

15. PROJECT RESULTS

Pocket Smart AI provides a working web application with authentication, multiple planning modules, recommendation generation, database support, history and a responsive user interface.

The project source code and phase-wise project documentation are maintained in the GitHub repository.

16. TEAM MEMBERS AND CONTRIBUTIONS

Ranjith P - Team Leader: project coordination and overall integration; backend/FastAPI development; database and application structure; AI integration; planner and recommendation implementation; GitHub repository management; overall testing and final integration.

Mohammed Reyan M: demo video preparation, application demonstration flow, demo recording and presentation support.

Shyam: PowerPoint presentation preparation, presentation structure and slide organization, presentation/demo support.

Magesh L: project documentation preparation, documentation organization and report support, documentation review.

17. PROJECT DOCUMENTATION AND DEMONSTRATION

The project includes phase-wise documentation covering Brainstorming & Ideation, Requirement Analysis, Project Design, Project Planning, Project Development, Project Testing, Project Documentation and Project Demonstration.

The project presentation is included in the Phase-8 Project Demonstration materials.

Demo Video: https://drive.google.com/file/d/1BKcmXrvjD-hHQUIMrUrjXarsnOMJT_f0/view?usp=drivesdk

18. FUTURE ENHANCEMENTS

- More planning categories*
- More product and service data sources*
- Improved recommendation personalization*
- Deployment as a public web application*
- Improved analytics and user preferences*
- Mobile-friendly enhancements*
- Additional AI-powered planning features*

19. CONCLUSION

Pocket Smart AI demonstrates how a web application, database services and generative AI can be combined to create a practical budget and recommendation assistant. The project provides a structured workflow for planning home interiors, parties and jewelry requirements while maintaining user history and a modular backend architecture.

20. GITHUB REPOSITORY

<https://github.com/Ranjithbsc/PocketSmart-ai>